

Giridhar Pai

Cyber Security Student | SOC Analyst Aspirant | Blue Team | Digital Forensics

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Professional Summary

Cybersecurity student with practical experience in blue teaming, digital forensics, cloud security, SIEM, web security, and secure application development. Built and contributed to internship, freelance, and hackathon projects involving authentication, encryption, logging, role-based access control, vulnerability scanning, and secure deployment. Strong interest in SOC operations, threat detection, incident analysis, and defense-oriented systems.

Education

Yenepoya University

Jul 2023 – Jul 2026

BCA in Cyber Security, Ethical Hacking, and Digital Forensics with IBM

CGPA: 9.15

Specialized in cybersecurity fundamentals, ethical hacking, digital forensics, and applied defense workflows.

S N Moodbidri Polytechnic

Aug 2019 – May 2022

Diploma in Computer Science and Engineering

SVS Vidyagiri English Medium School, Bantwal

2007 – 2016

Primary and Higher Schooling

Technical Skills

- **Security Operations:** SIEM (Splunk), threat hunting, IDS/IPS concepts, incident response basics, vulnerability assessment, DDoS mitigation
- **Cloud Security:** AWS Security (IAM, Secrets Manager, secure architecture)
- **Forensics:** Digital forensics, metadata extraction, heatmap analysis
- **Networking:** Network fundamentals, CompTIA Network+

Tools & Technologies

- Splunk (Enterprise Security, SOAR), Linux, Docker, Git
- Python, Java, C, JavaScript, SQL, HTML/CSS

Certifications

- AWS Certified Security - Specialty
- Google Cybersecurity Specialization
- Digital Forensics Essentials (EC-Council)
- Managing Security and Administration in SAP Analytics Cloud
- Splunk the Complete Course from Zero to Hero 2022
- Splunk Training - 10 Courses
- CompTIA Network+
- Deloitte Australia Cyber Job Simulation Internship
- TryHackMe - 13 Certifications

Experience and Projects

Tejas Raksha – Automated Web Application Security Scanner

Mar 2026 – Apr 2026

- Engineered a multi-tenant web security scanner for agricultural portals to identify 12+ OWASP Top 10 vulnerability categories, including SQL injection, XSS, SSRF, and command injection.
- Built a distributed microservices platform using FastAPI, Celery, Redis, and PostgreSQL.
- Developed a responsive frontend with React, Vite, and Tailwind CSS, featuring real-time scan progress updates through WebSockets and PDF report generation.
- Containerized services with Docker Compose and created a plugin-based scanning workflow with false-positive reduction stages.
- Added scheduled scanning, trend analysis, CVE enrichment, and CI/CD integration with Jira, Slack, and email.

Kids Wonder Learn (KWL) – Freelance Project

2025 – 2026

- Built a browser-based educational platform for children aged 2–13+ with student, parent, teacher, and admin roles.
- Developed interactive learning modules including games, dashboards, progress tracking, activity logging, and secure access workflows.
- Used HTML5, CSS3, JavaScript, Supabase, and Netlify to design a modular and scalable front-end architecture.
- Planned secure backend integration with authentication, PostgreSQL schema design, role mapping, and access control.

Manufacturing Vault – Freelance Project

Mar 2026 – Apr 2026

- Built a production-grade file encryption platform using AES-256-GCM and Argon2id key derivation with a zero-persistence architecture.
- Deployed on Azure using Purdue Model-aligned 4-tier VNet segmentation and Azure WAF v2 for perimeter defense.
- Achieved a 100% pass rate on a 45-point security audit covering cryptography, network isolation, and operational controls.

Kinetic Forge – Freelance Project

Mar 2026 – Apr 2026

- Implemented TOTP-based MFA, Google OAuth 2.0 SSO, and role-based access control for supply chain users.
- Encrypted sensitive cargo telemetry, built a real-time shipment map, and created tamper-resistant audit logging.
- Defended against SQLi, XSS, CSRF, and brute-force attacks through rate limiting and lockout controls.

Vault Protocol – Freelance Project

Mar 2026 – Apr 2026

- Designed a dual-authorization workflow for financial transactions with mandatory MFA re-authentication.
- Implemented idempotency keys using UUIDv7 and Redis TTL to prevent duplicate financial actions.
- Added encrypted PII storage, scrypt-based password hashing, and hash-chained audit logging.

Retail Data Security – Freelance Project

Mar 2026 – Apr 2026

- Architected a three-layer defense model combining Azure WAF v2, VNet isolation, and AES-256-GCM encryption.
- Implemented Argon2id-based key derivation, JWT authentication, Redis-cached keys, and encrypted blob storage.
- Validated the system using tamper tests and web attack resistance checks.

Obsidian Pulse – Freelance Project

Mar 2026 – Apr 2026

- Developed a healthcare security platform with dual-factor identity restoration, encrypted PHI handling, and real-time telemetry.
- Added brute-force resistance through quarantine logic, Redis-backed idempotent actions, IP blacklisting, and audit logging.
- Built for secure and latency-aware healthcare workflows.

EdTech Security Framework – Freelance Project

Mar 2026 – Apr 2026

- Implemented FIDO2/WebAuthn passkeys, TOTP fallback, and Google OAuth 2.0 SSO with defense-in-depth controls.
- Added restoration logic, login lockouts, IP blacklisting, encrypted secrets, and tamper-resistant audit verification.
- Designed to demonstrate secure passwordless authentication and audit integrity for educational systems.

Projects

- **IBM Internship – Tejas Raksha:** Zero-day scanner for agricultural web portals; identified OWASP Top 10 weaknesses and proposed secure coding fixes.
- **DefendX:** Real-time threat detection and monitoring system simulating enterprise SOC operations; visualized alerts and security events for incident response.
- **AI-Powered Deepfake Detection System:** Deepfake detection using heatmap analysis and metadata extraction; reported 99% accuracy in hackathon work.
- **Password Strength Checker:** Client-side tool evaluating password entropy and patterns to promote secure password practices.
- **Kids Wonder Learn (KWL):** Educational platform with role-based access, activity logging, and security-focused configuration; deployed in offline environment.
- **Doctors Appointment Booking System:** Award-winning healthcare workflow system; implemented authentication, access control, and offline deployment.

Hackathons and Competitions

- CIDECODE Hackathon (2025) – Consolation Prize Winner for DeepFake Detection System
- IBM ICE DAY 2025 – 2nd Place in CTF Competition and 2nd Runner-Up in 24-hour Hackathon
- CODE SPRINT – Finalist (Top 7 out of 65 teams) and Consolation Prize Winner
- TryHackMe – Ranked in the Top 1% globally

Achievements

- Completed 23+ certifications and security-related learning milestones across cloud, SIEM, forensics, and cyber defense.
- Built practical experience through labs, CTFs, simulations, and hands-on tooling.
- Actively contributed to cybersecurity communities and student initiatives.

Leadership / Community

- Member, OWASP Yenepoya

Apr 2024 – Present

- Participated in security challenges, cybersecurity research, and community activities.

Publications / Writing

- The Aggressive Defender: A Technical Guide to Proactive Threat Hunting
- The Thin Digital Line: Why India's Defense Now Rests in Your Hands
- OWASP Yenepoya Hackers Unveiled post and related cybersecurity reflections

Additional Links

- GitHub: github.com/Giri227
- LinkedIn: linkedin.com/in/giridharpai227/
- Portfolio: giri227.github.io/Giridharpai/